

Clean Synthetic Core Medicine Search Benchmark

The 66,257-case primary test file, eleven retrieval systems, and failure evidence

medicine-search-clean engineering analysis, 23 July 2026

Result in one paragraph. The main denominator is now **66,257** distinct, collision-free, positive-retrieval query-target pairs from `data/05_synthetic_clean_core/test_cases.csv`. Every system receives the same case IDs and the verified target remains hidden until scoring. Algorithm 5 reaches **98.2055%** Hit@1 and **100.0000%** Hit@20. The table results below are newly computed on this file; no score from the 595-row OCR or 115,000-row inclusive benchmark is reused.

1 Main Testing Contract

Stage	Concrete example	What the benchmark does
Input	0aflon	Pass the corrupted commercial-name text to the search system.
Hidden answer	DAFLON	Keep the verified family outside retrieval; use it only after ranking.
Search output	ranked family names	Search the same 25,066-row runtime catalog with a limit of 20 families.
Scoring	Hit@1, Hit@5, Hit@10, Hit@20, MRR@20	Locate the first acceptable family and calculate rank and safety metrics.
Primary vote	one case ID	Count each compact query-target pair once, regardless of source repetition.

Table 1: Input, process, and output for every row. The expected family never enters candidate generation.

Audit item	Value	Meaning
Original generated rows	115,000	Complete 34-category source benchmark.
Source rows represented in clean core	66,716	Rows retained before duplicate pair collapse.
Source rows outside clean core	48,284	Behavioral, ambiguous, exact, no-match, or unsupported rows not used in this positive-retrieval denominator.
Primary clean unique pairs	66,257	Every row receives one vote in the headline score.
Collapsed duplicate occurrences	459	Repeated query-target occurrences removed from weighting.
Unique corrupted queries	66,257	No compact query is repeated.
Verified target families	16,845	All targets resolve to the 25,066-row runtime catalog.
Clean category labels	22	Membership labels; rare rows can carry more than one.
Clean error-type labels	4,190	Exact mutation labels after corrections.
Rows with a cleaning note	17,230	Rows relabelled or retained after a documented safety comparison.
Dangerous gold-closer pairs	158	Gold target is closer than the recorded alternative; both remain visible for safety analysis.

Table 2: The clean-core CSV is the headline denominator. Counts reconcile from the 115,000-row source to retained source occurrences and then to unique primary pairs.

Check	Observed result	Why it matters
Source identity	commit 47706ad; SHA-256 65f81b58dee1e7127386652383d2f6a8db1734e832dcbc73f14a9b831678886f	The local CSV is byte-identical to the supplied clean-core file.
Case identity	66,257 unique IDs; 66,257 unique compact queries	No row is counted twice.
Target coverage	16,845 unique verified families; zero unmapped targets	A retrieval miss cannot be caused by an absent expected family.
Expected behavior	all rows are <code>match</code> and <code>use_in_primary_score=1</code>	Hit@k has one unambiguous positive-retrieval meaning.
Catalog status	all rows say <code>product_scope_unverified</code>	Product-scope validation remains a limitation even though every family key resolves.
Split	target-family hash, 80% development and 20% holdout	No target family occurs in both diagnostic splits.

Table 3: Validation gates applied before any algorithm score was accepted.

2 Change From The 115,000-Row File

The old file mixed positive typo retrieval with exact matches, no-match cases, ambiguity behavior, route/form contradictions, and stress generators. The clean core keeps only positive-retrieval rows with defensible text evidence, corrects the effective mutation label, and collapses repeated compact query-target pairs.

Transformation	Before	After	Difference
Generated source rows	115,000	66,716	48,284 removed
Retained source occurrences	66,716	66,257	459 duplicate occurrences collapsed
Primary scoring rows	109,974 in the earlier fair view	66,257	New task-specific denominator
Exact real-name collision rows	5,026 in the earlier fair audit	0	Excluded before this core was formed

Table 4: The clean core is not a filtered score view over 115,000 rows. It is a new unique-pair positive-retrieval benchmark.

System	Old fair H@1	Clean H@1	Δ H@1	Old fair H@20	Clean H@20	Δ H@20
Algorithm 1, current app	78.6841%	83.8583%	+5.1743 pp	90.6578%	92.2333%	+1.5755 pp
Algorithm 2, external fast	82.9251%	92.4053%	+9.4803 pp	93.9740%	96.8788%	+2.9048 pp
Algorithm 3, rank fusion	84.7373%	91.8363%	+7.0990 pp	95.3389%	97.1203%	+1.7814 pp
Algorithm 4, family rescue	85.8694%	97.1852%	+11.3158 pp	95.6490%	99.5155%	+3.8666 pp

Table 5: Algorithms 1–4 before and after changing the denominator. “Old fair” uses 109,974 collision-excluded generated rows; “clean” uses 66,257 unique positive-retrieval pairs. Delta measures dataset and task composition, not a code change.

Table 6: Original category coverage after clean-core construction. A clean case can retain more than one provenance label, so membership counts are diagnostic rather than another denominator.

Original category	Source rows	Clean memberships	Role in the clean core
1. Single Letter Visual Confusion	8,000	7,841	Core corruption retained
2. Single Letter Phonetic Confusion	6,000	5,983	Core corruption retained
3. Multi Char Phonetic Confusion	5,000	4,991	Core corruption retained
4. Ligature Confusion	4,000	3,992	Core corruption retained
5. Single Char Deletion Position Weighted	5,000	4,966	Core corruption retained
6. Single Char Insertion	3,000	2,988	Core corruption retained
7. Transposition Position Weighted	3,000	3,000	Core corruption retained
8. Truncation Doctor Abbreviation	6,000	0	Absent from positive-retrieval core
9. Two Error Combinations	8,000	7,926	Core corruption retained
10. Three Error Combinations	5,000	4,958	Core corruption retained
11. Speed Typing Errors	3,000	2,981	Core corruption retained
12. Wrong Vowels In Consonant Frame	5,000	4,870	Core corruption retained
13. Punctuation Whitespace Copy Paste Artifacts	1,000	0	Absent from positive-retrieval core
14. Case Sensitivity	500	0	Absent from positive-retrieval core
15. Number Word Confusion	500	0	Absent from positive-retrieval core
16. Embedded Form Strength Parsing	4,000	0	Absent from positive-retrieval core
17. Dangerous ED1 Pairs	5,000	158	Core corruption retained
18. Substring Traps	3,000	0	Absent from positive-retrieval core
19. Negative No Match Expected	3,000	0	Absent from positive-retrieval core
20. Contradictory Form Route	2,000	0	Absent from positive-retrieval core
21. Cancelled Na Drug Lookup	1,000	0	Absent from positive-retrieval core
22. Score Gap Ambiguity Detection	4,000	0	Absent from positive-retrieval core
23. OCR Letter Digit Confusion	4,000	3,998	Core corruption retained
24. OCR + Other Error Combined	3,000	2,999	Core corruption retained
25. Double Letter Reduction Expansion	3,000	2,987	Core corruption retained
26. Keyboard Adjacent Sampled	2,000	1,924	Core corruption retained
27. Four + Error Combinations	4,000	7	Small stress subset retained
28. Consonant Frame Wrong Vowels Heavy	4,000	68	Small stress subset retained
29. Multi Word Name Fragmentation	2,000	0	Absent from positive-retrieval core
30. Autocorrect Artifacts	2,000	0	Absent from positive-retrieval core
31. Exact Match Baseline	2,000	0	Absent from positive-retrieval core
32. Exact Match With Strength	2,000	0	Absent from positive-retrieval core
33. Keyboard Shift Whole Word	500	0	Absent from positive-retrieval core
34. Prefix Ambiguity Awareness	1,500	0	Absent from positive-retrieval core

Cleaning action	Rows	Share	Concrete interpretation
No relabelling needed	49,027	74.00%	Original label agreed with the observed final mutation.
combined-operation case relabelled by final effective Damerau distance	15,851	23.92%	The final string, rather than the planned chain length, sets the clean category.
declared subtype corrected from 3 to 2 actual vowel changes	540	0.82%	Two vowels changed even though the generator subtype said three.
declared subtype corrected from 3 to 1 actual vowel changes	371	0.56%	One vowel changed even though the generator subtype said three.
declared subtype corrected from 2 to 1 actual vowel changes	296	0.45%	One vowel changed even though the generator subtype said two.
gold distance 1; closest competitor distance 2	119	0.18%	The dangerous pair stays scoreable because the gold family is strictly closer.
combined-operation case relabelled by final effective Damerau distance; gold distance 1; closest competitor distance 2	16	0.02%	Several documented corrections apply to the same row.
declared subtype corrected from 2 to 1 actual vowel changes; declared subtype corrected from 3 to 1 actual vowel changes	14	0.02%	Several documented corrections apply to the same row.

Table 7: Cleaning notes explain every relabelled or safety-retained row. Blank notes mean the original mutation label already matched the observed corruption.

3 Dataset Composition

Table 8: The dominant denominator contains one- and two-edit corruptions. Distance four and five remain as a small severe tail rather than dominating the score.

Dimension	Group	Cases	Share	Plain meaning
Effective Levenshtein	1	38,664	58.35%	Ordinary insertion, deletion, or replacement count.
Effective Levenshtein	2	20,248	30.56%	Ordinary insertion, deletion, or replacement count.
Effective Levenshtein	3	6,611	9.98%	Ordinary insertion, deletion, or replacement count.
Effective Levenshtein	4	421	0.64%	Ordinary insertion, deletion, or replacement count.
Effective Levenshtein	5	313	0.47%	Ordinary insertion, deletion, or replacement count.
Effective Damerau	1	41,665	62.88%	Adjacent transposition counts as one edit.
Effective Damerau	2	18,335	27.67%	Adjacent transposition counts as one edit.
Effective Damerau	3	5,694	8.59%	Adjacent transposition counts as one edit.
Effective Damerau	4	563	0.85%	Adjacent transposition counts as one edit.
Difficulty	Easy	586	0.88%	Generator-assigned challenge level.
Difficulty	Extreme	17,913	27.04%	Generator-assigned challenge level.
Difficulty	Hard	33,906	51.17%	Generator-assigned challenge level.
Difficulty	Medium	13,852	20.91%	Generator-assigned challenge level.

Dimension	Group	Cases	Share	Plain meaning
Danger	Dangerous	158	0.24%	Safety label for a close named competitor.
Danger	Safe	66,099	99.76%	Safety label for a close named competitor.
Shared bigrams	0 Shared Bigrams	349	0.53%	Adjacent two-character anchors retained in the target.
Shared bigrams	1 Shared Bigram	1,144	1.73%	Adjacent two-character anchors retained in the target.
Shared bigrams	2 3 Shared Bigrams	6,741	10.17%	Adjacent two-character anchors retained in the target.
Shared bigrams	4 + Shared Bigrams	58,023	87.57%	Adjacent two-character anchors retained in the target.

The source label `difficulty=EXTREME` contains 17,913 generator-design cases. It is not the normalized distance cohort: the later `r>0.60` rule contains 123 cases.

Table 9: Clean mutation-category membership with one actual row-level example per category.

Clean category	Cases	Share	Example input	Expected family	Edits
Single Letter Visual Confusion	7,841	11.83%	happinessbarylfe	HAPPINESSBABYLIFE	1
Combined Errors Effective 2	7,463	11.26%	undaktoven	INDACTOVEN	2
Single Letter Phonetic Confusion	5,983	9.03%	viratinantiseptic	VIRADINANTISEPTIC	1
Multi Char Phonetic Confusion	4,991	7.53%	kaminomotohairgrowthaccele ratoryiupgrade	KAMINOMOTOHAIRGROWTHACCELE RATORIIUPGRADE	1
Single Char Deletion Position Weighted	4,966	7.50%	mutelaskinfreshener	MUSTELASKINFRESHENER	1
Wrong Vowels Relabelled By Actual Changes	4,870	7.35%	cupralex	CIPRALEX	1
Combined Errors Effective 3	4,346	6.56%	qiabtivia	GLAPTIVIA	3
OCR Letter Digit Confusion	3,998	6.03%	imipenemcilastatinkab1	IMIPENEMCILASTATINKABI	1
Ligature Confusion	3,992	6.03%	touliflyfacal	TOULIFLYFACIAL	1
Transposition Position Weighted	3,000	4.53%	chaplha	CHALPHA	2
OCR + Error Effective 2	2,997	4.52%	vitamlnkmeniq7	VITAMINKMENAQ7	2
Single Char Insertion	2,988	4.51%	calciumviit	CALCIUMVIT	1
Double Letter Reduction Expansion	2,987	4.51%	ssimvacor	SIMVACOR	1
Speed Typing Errors	2,981	4.50%	rngeracetateelnasr	RINGERACETATEELNASR	1
Keyboard Adjacent Sampled	1,924	2.90%	billakordhair	VILLAKORDHAIR	1
Combined Errors Effective 4 +	563	0.85%	myclhaeme	MEDAHEME	4
Combined Errors Effective 1	496	0.75%	whivewello	WHITEWELLO	1
Dangerous Name Pair Gold Closer	158	0.24%	casty	CASTO	1
Consonant Frame Wrong Vowels Heavy	68	0.10%	nopyl	NOPEL	1
Extreme Combined Effective 2	4	0.01%	vfnt	VFEND	2
Extreme Combined Effective 3	3	0.00%	mokce	MEGACE	3
OCR + Error Effective 1	2	0.00%	ope	OPIE	1

Table 10: Most frequent exact mutation labels. The complete machine-readable metric file contains all 4,190 labels.

Clean error type	Cases	Share	Example input	Expected family
Wrong 1 Vowel Changes	2,299	3.47%	cupralex	CIPRALEX
Wrong 2 Vowel Changes	1,824	2.75%	tobrialyx	TOBRAALEX
Wrong 3 Vowel Changes	747	1.13%	metuniemevarydayeasy	METANIUMEVERYDAYEASY
Double First Letter	645	0.97%	vvalproexcr	VALPROEXCR
Stutter Prefix 2	617	0.93%	chchampix	CHAMPIX
Repeat Last Letter	576	0.87%	unasynn	UNASYN
Stutter Prefix 3	576	0.87%	bosbostonximajointh	BOSTONXIMAJOINTH
Skip Second Letter	567	0.86%	rngeracetateelnasr	RINGERACETATEELNASR
Deletion First Char Pos 0	496	0.75%	ustelagentlecleansing	MUSTELAGENTLECLEANSING
Vowel + Phonetic Variant 3 Effective Damerau 2	435	0.66%	undaktoven	INDACTOVEN
Deletion + Phonetic Variant 1 Effective Damerau 2	424	0.64%	darbrim	DARAPRIM
Deletion + Phonetic Variant 3 Effective Damerau 2	410	0.62%	isiskingerlmonfilterbags	ISISGINGERLEMONFILTERBAGS
Vowel + Phonetic Variant 0 Effective Damerau 2	404	0.61%	polunehair	BOLINEHAIR
Vowel + Phonetic Variant 2 Effective Damerau 2	404	0.61%	muxidrix	MOXITRIX
Visual + Phonetic Variant 1 Effective Damerau 2	387	0.58%	phoztec	PROSTEC
Deletion + Phonetic Variant 2 Effective Damerau 2	384	0.58%	rotakrbamide	ROTACARBAMIDE
Vowel + Phonetic Variant 1 Effective Damerau 2	380	0.57%	antocholyst	ANDOCHOLEST
Transposition + Deletion Variant 0 Effective Damerau 2	378	0.57%	igvarrock	GIVAROROCK
Visual + Phonetic Variant 3 Effective Damerau 2	374	0.56%	carhitopalevodopaeva	CARBIDOPALEVODOPAeva
Deletion + Phonetic Variant 0 Effective Damerau 2	370	0.56%	trbsiflo	TRYPSIFLO
Transposition + Deletion Variant 1 Effective Damerau 2	357	0.54%	craviprss	CARVIPRESS
Visual Visual Phonetic Variant 1 Effective Damerau 3	344	0.52%	dhlxy	TRIXY
Visual + Phonetic Variant 2 Effective Damerau 2	343	0.52%	zynorent	ZYMOREND
Visual Visual Phonetic Variant 2 Effective Damerau 3	329	0.50%	icurtra	ICANDRA
Vowel Phonetic Deletion Variant 2 Effective Damerau 3	329	0.50%	arckhuma	ARCECHIMA

4 Methods And Metrics

Table 11: The fixed comparison roster. Every row uses the same catalog, case IDs, top-20 cutoff, and scoring code.

Order	System	Evidence used
1	Exact or prefix match	Exact compact family, then families that begin with the compact query.
2	Exhaustive Levenshtein	Smallest insert, delete, or replace count over every family name.

Order	System	Evidence used
3	Jaro-Winkler	Character-order similarity with an explicit shared-prefix reward.
4	Character 3-gram TF-IDF	Cosine similarity between weighted three-character pieces.
5	RapidFuzz token ratio	Token-set and token-sort string similarity, keeping the larger score.
6	Phonetic baseline	Medicine phonetic key, then Levenshtein ranking between sound keys.
7	Algorithm 1	Current application retrieval and safety ranking.
8	Algorithm 2	Fast indexed exact, edge, n-gram, skeleton, phonetic, and delete-key retrieval.
9	Algorithm 3	Weighted rank fusion of Algorithms 1 and 2.
10	Algorithm 4	Algorithm 2 plus family rescue, multi-signal scoring, bounded correction, and clarification.
11	Algorithm 5	Evidence-guided retrieval, mixed-error expansion, bounded reranking, and clarification.

4.1 Algorithm 5 Score Signals

Algorithm 5 adds applicable signals rather than stopping at the first matching rule. A weight is the signal’s maximum score contribution, not a guarantee that the candidate wins.

Table 12: Every main signal in the Algorithm 5 rescue score.

Signal	Weight	Plain-English meaning	Example
Exact family	1.25	Compact query exactly equals a catalog family.	RIVOTRIL→RIVOTRIL. Exact evidence is strong; an exact different real drug is excluded from this clean core.
Ordinary edit similarity	0.58	Rewards fewer ordinary edits after accounting for string length.	Both CEFAXIM and CEFIXIME are one edit from CEFAXIME, so this signal ties.
Weighted edit similarity	0.52	Gives lower cost to known confusion pairs and vowel changes. Known pairs cost 0.45, vowel changes 0.70, and an ordinary replacement 1.	For CEFAXIME, CEFIXIME has weighted distance 0.70 while CEFAXIM has 1.00.
Same-position evidence	0.24	Measures query characters retained at the same indexes.	CEFAXIME and CEFIXIME match in seven of eight positions, producing 0.875.
Character 3-grams	0.18	Compares overlapping three-character pieces.	RIVOTIL and RIVOTRIL share RIV, IVO, and VOT.
Prefix evidence	0.16	Measures uninterrupted matching characters from the beginning.	CLARANE and CLARA share CLARA; this can favor a shorter wrong family.
Consonant skeleton	0.16	Compares a reduced key that preserves consonant structure.	NOW and NEW preserve NW; the key cannot decide which vowel was seen.
Length coverage	0.16	Rewards candidate length close to query length.	Eight-character CEFIXIME has full coverage; seven-character CEFAXIM has $7/8 = 0.875$.
Phonetic key	0.14	Compares sound-confusion groups such as C/K/Q, F/V, P/B, D/T, and S/Z.	COUFSED retains sound evidence for COUGHSED, but still needs lexical support.
Suffix evidence	0.10	Measures uninterrupted matching characters from the end.	ANYOLAX and MYOLAX share YOLAX.

Signal	Weight	Plain-English meaning	Example
Character 2-grams	0.10	Compares adjacent two-character pieces.	MYMLAX and MYOLAX retain several pairs despite one replacement.
Ordered subsequence	0.10	Rewards query letters that occur in candidate order despite gaps.	RIVOTRI is an ordered subsequence of RIVOTRIL.

4.2 Bonuses And Penalties

Table 13: Bonuses require supporting conditions. No single bonus is an automatic winner.

Condition	Change	Meaning	Specific word example
Same first character	+0.12	Useful evidence, too small to become a universal first-character rule.	CEFAXIME and CEFIXIME both start with C.
Confusable first characters	+0.06	Retains a candidate when first letters form a known pair such as P/B or C/K.	BANDOL→PANDOL receives the smaller B/P bonus.
Length difference at most 1 and weighted similarity at least 0.76	+0.18	Rewards a compact near-match only when spelling is already strong.	CEFAXIME→CEFIXIME has equal length and weighted similarity 0.9125.
Prefix at least 0.55 and weighted similarity at least 0.66	+0.08	Rewards a strong beginning only when full spelling remains plausible.	CLARANE→CLARA has prefix $5/7 = 0.714$.
Weighted similarity at least 0.84 and position at least 0.70	+0.22	Rewards agreement between independent spelling measurements.	CEFAXIME→CEFIXIME passes both thresholds.
Short partial prefix with weak weighted evidence	−0.18 to −0.42	Stops a short prefix from defeating a stronger complete name.	ABIMOL→ABIMOLEXTRA can receive the maximum penalty.
Very short non-exact query with weak edit evidence	−0.22	Reduces confidence from three- or four-character noise.	ABC→ABCIMMUNE contains too little full-name evidence.
Weak spelling plus weak skeleton or phonetic evidence	−0.20	Penalizes candidates supported by no strong lexical or sound channel.	GHONT→CALAMINE has no strong support.
Weak prefix, weak 3-grams, and weak edit evidence	−0.10	Penalizes little shared local structure.	GHONT→CALAMINE also lacks a prefix and trigrams.

4.3 Bounded Corrections

Table 14: Except for controlled pure deletion, equal ordinary distance does not activate a correction.

Correction	Required evidence	Why the bound exists	Specific word example
Strictly closer candidate	Both retrieval layers find the alternative; distance at most 2 and at least one edit closer; score gap at most 0.25.	Agreement and the gap limit the change to a small inversion.	MYCLAX: supported MYOLAX can replace a distance-two first result.
Pure deletion candidate	Query length at least 5; ordered subsequence; one or two missing characters; no farther than current first; score gap at most 0.40.	Every visible character must stay in order. This is the only correction allowed at equal raw distance.	RIVOTRI→RIVOTRIL is missing only the final L.
Single-token correction	Current first has several words; alternative has one; alternative is strictly closer within distance 2; gap at most 0.20.	A short family cannot win only because it is shorter.	ABIMOL may replace ABIMOLEXTRA when the full evidence is close.
Family-head correction	Catalog confirms a variant; query length at least 5; head within two edits and strictly closer; gap at most 1.40.	Catalog grouping blocks arbitrary prefixes from claiming family membership.	LANTS can recover the validated LANTUS variant family.
Strict full-name correction	Query length at least 5; rescue-only name within three edits and strictly closer; weighted distance no worse; gap at most 0.35; no protected current evidence.	The alternative must win on raw and weighted spelling without defeating independent protected evidence.	MYOLANA can move MYOLAX from rank 2 to rank 1.

Metric	Calculation	Meaning
Hit@1	target family occurs at rank 1	The first displayed family is correct.
Hit@20	target family occurs at ranks 1 through 20	Retrieval found a usable candidate.
MRR@20	$1/r$ for first target rank $r \leq 20$, otherwise 0	Earlier correct candidates receive more credit.
Failure rate	misses in group divided by rows in group	Finds the weakest group.
Failure share	misses in group divided by all misses for that system	Finds the group causing the most total damage.
Unsafe confident top-1	wrong first family shown as confident without clarification	Safety failure; rank alone cannot determine it.
Latency	warm query wall time after index preparation	Implementation time on this machine, not asymptotic complexity.

Table 15: Metric definitions used in every later table. Percentages never mix denominators.

5 Overall Results

System	H@1	H@5	H@10	H@20	MRR@20	Build ms	Median ms	P95 ms	Mean candidates	Unsafe
Exact or prefix match	3.0019%	3.0125%	3.0125%	3.0125%	0.030072	1.6	0.7079	0.8049	0.06	0.0000%
Exhaustive Levenshtein	94.0489%	98.3956%	99.0250%	99.4416%	0.959861	0.0	0.7357	0.7990	17476.00	0.0000%
Jaro-Winkler	89.5694%	96.2630%	97.5459%	98.4273%	0.925647	0.0	0.7249	0.7907	17476.00	0.0000%
Character 3-gram TF-IDF	75.2071%	86.7154%	89.4230%	91.6613%	0.804208	64.0	0.5126	0.6210	921.27	0.0000%
RapidFuzz token ratio	65.6716%	78.2498%	82.0774%	85.4717%	0.713468	0.4	8.1414	12.2144	17476.00	0.0000%
Phonetic baseline	69.6560%	84.9223%	89.0396%	92.0703%	0.764241	0.2	0.6924	0.7842	17475.74	0.0000%
Algorithm 1, current app	83.8583%	89.7188%	91.2507%	92.2333%	0.863662	4722.1	1.2253	4.5748	35.91	0.0000%
Algorithm 2, external fast	92.4053%	96.3325%	96.7324%	96.8788%	0.941355	4344.6	5.7286	21.9070	93.77	0.1675%
Algorithm 3, rank fusion	91.8363%	96.1830%	96.7309%	97.1203%	0.938051	12792.4	8.3464	24.4607	34.78	0.0000%
Algorithm 4, family rescue	97.1852%	99.2423%	99.4250%	99.5155%	0.980944	8737.0	14.6203	31.5124	24.36	0.0000%
Algorithm 5, evidence-guided rescue	98.2055%	99.6619%	99.8989%	100.0000%	0.988569	16750.6	52.1872	117.7635	24.54	0.0000%

Table 16: Primary performance over all 66,257 clean unique pairs. Build time is one-time index preparation; median and P95 are warm query times.

Accuracy rows use the same cases and scoring code. The refreshed Algorithm 5 latency was collected with eight parallel workers, while the retained baseline latencies came from single-process runs. Those timing columns document each artifact, but they are not a controlled speed comparison.

1. Algorithm 5, evidence-guided rescue has the highest Hit@1 at 98.2055%; Algorithm 5, evidence-guided rescue has the highest Hit@20 at 100.0000%.
2. Relative to exhaustive Levenshtein, Algorithm 5 changes Hit@1 by +4.1565 pp and Hit@20 by +0.5584 pp.

System	Clarification rate	No-result rate	Unsafe confident top-1	Mean candidate count
Exact or prefix match	3.22%	96.78%	0.0000%	0.06
Exhaustive Levenshtein	100.00%	0.00%	0.0000%	17476.00
Jaro-Winkler	100.00%	0.00%	0.0000%	17476.00
Character 3-gram TF-IDF	99.71%	0.29%	0.0000%	921.27
RapidFuzz token ratio	100.00%	0.00%	0.0000%	17476.00
Phonetic baseline	100.00%	0.00%	0.0000%	17475.74
Algorithm 1, current app	98.06%	1.94%	0.0000%	35.91
Algorithm 2, external fast	55.47%	0.15%	0.1675%	93.77
Algorithm 3, rank fusion	99.96%	0.04%	0.0000%	34.78
Algorithm 4, family rescue	99.95%	0.05%	0.0000%	24.36
Algorithm 5, evidence-guided rescue	100.00%	0.00%	0.0000%	24.54

Table 17: Behavior uses response metadata, not rank alone. Mean candidate count is each algorithm’s exposed considered or matched pool, not the common top-20 output size.

System compared with A5	$\Delta H@1$	A5-only H@1	Other-only H@1	p H@1	$\Delta H@20$	p H@20
Exact or prefix match	+95.2035 pp	63,079	0	$< 10^{-300}$	+96.9875 pp	$< 10^{-300}$
Exhaustive Levenshtein	+4.1565 pp	2,923	169	$< 10^{-300}$	+0.5584 pp	8.32×10^{-112}
Jaro-Winkler	+8.6361 pp	5,870	148	$< 10^{-300}$	+1.5727 pp	4.24×10^{-314}
Character 3-gram TF-IDF	+22.9983 pp	15,304	66	$< 10^{-300}$	+8.3387 pp	$< 10^{-300}$
RapidFuzz token ratio	+32.5339 pp	21,691	135	$< 10^{-300}$	+14.5283 pp	$< 10^{-300}$
Phonetic baseline	+28.5494 pp	19,065	149	$< 10^{-300}$	+7.9297 pp	$< 10^{-300}$
Algorithm 1, current app	+14.3472 pp	9,644	138	$< 10^{-300}$	+7.7667 pp	$< 10^{-300}$
Algorithm 2, external fast	+5.8001 pp	4,046	203	$< 10^{-300}$	+3.1212 pp	$< 10^{-300}$
Algorithm 3, rank fusion	+6.3691 pp	4,417	197	$< 10^{-300}$	+2.8797 pp	$< 10^{-300}$
Algorithm 4, family rescue	+1.0203 pp	688	12	1.02×10^{-185}	+0.4845 pp	4.68×10^{-97}

Table 18: Paired case-level comparison. Delta is A5 minus the other system; exact McNemar p tests whether discordant wins are balanced.

System	Dev. H@1	Holdout H@1	Dev. H@20	Holdout H@20
Exact or prefix match	2.95%	3.21%	2.96%	3.22%
Exhaustive Levenshtein	94.17%	93.57%	99.45%	99.42%
Jaro-Winkler	89.62%	89.38%	98.43%	98.43%
Character 3-gram TF-IDF	75.22%	75.15%	91.78%	91.18%
RapidFuzz token ratio	65.95%	64.56%	85.43%	85.63%
Phonetic baseline	69.52%	70.18%	92.08%	92.04%
Algorithm 1, current app	83.97%	83.42%	92.30%	91.97%
Algorithm 2, external fast	92.51%	91.99%	96.96%	96.56%
Algorithm 3, rank fusion	91.94%	91.44%	97.22%	96.73%
Algorithm 4, family rescue	97.24%	96.96%	99.53%	99.47%
Algorithm 5, evidence-guided rescue	98.22%	98.16%	100.00%	100.00%

Table 19: Target-family-disjoint diagnostic split. The all-case score remains primary; this table checks whether rankings transfer to held-out families.

6 Deep Audit Of The Remaining Rank-One Errors

The strict pair-level score counts the source family as correct even when the corrupted text is equally close or closer to another real catalog family. The tables below preserve that score, then separate ranking defects, retrieval defects, and text-level ambiguity.

Mutually exclusive outcome	Cases	Share of 66,257
Correct at rank 1	65,068	98.2055%
Expected at ranks 2–5	965	1.4564%
Expected at ranks 6–10	157	0.2370%
Expected at ranks 11–20	67	0.1011%
Expected outside top 20	0	0.0000%

Table 20: The arithmetic behind 98.2055% Hit@1. The 1,189 first-rank misses contain 1,189 ranking failures, where the expected family is still in the top 20, and 0 retrieval failures. Only 0 queries return no candidates.

Independent check	Observed	Acceptance rule
A5 rows	66,257	Must equal the clean-core denominator.
Unique A5 case IDs	66,257	One result row per test case.
Recomputed Hit@1 mismatches	0	Top name was compared again with the expected family key.
Recomputed Hit@20 mismatches	0	Stored rank and Hit@20 flag must agree.
Levenshtein mismatches	0	Every stored distance was recalculated from query and target.
Unrestricted Damerau mismatches	0	The stored effective-Damerau field was recalculated with its own distance definition.
Stored Damerau versus A5 OSA differences	88	These are expected definition differences, not Hit@k scoring errors.
Exact different-family queries	0	A positive value would be an unfair real-name collision.
Unique-nearest failures rerun	14	Current A5 was executed again with candidate scores exposed.
Unique-nearest rerun mismatches	0	Fresh top name and expected rank must match the benchmark.
A5/OSA distance test pairs	10,000	Random string pairs used to verify the distance implementation.
A5/OSA implementation mismatches	0	A positive value would invalidate OSA geometry labels.

Table 21: The score was recomputed from row-level names and ranks. IDs, Hit@k flags, Levenshtein values, unrestricted Damerau values, and exact-collision checks reproduce exactly. The 88 stored-Damerau versus A5-OSA differences are expected because those fields use different distance definitions; neither field determines Hit@k.

The clean-core column **effective_damerau** uses unrestricted Damerau–Levenshtein distance. Algorithm 5’s raw edit feature uses OSA, which does not permit a character substring to be edited more than once. For example, **altsren**→**ALSOTRIN** has stored unrestricted-Damerau distance 3 but A5 OSA distance 4. This definition difference occurs in 88 of 66,257 rows. It does not change the benchmark answer or the rank-based Hit@1 calculation.

Denominator	Units	Hit@1	Hit@20	What receives one vote
Pair micro-average	66,257	98.2055%	100.0000%	Every distinct query-target pair receives one vote.
Family macro-average	16,845	97.5120%	100.0000%	Every verified family receives one vote after averaging its cases.

Table 22: The headline is the pair micro-average. Family balancing lowers Hit@1 to 97.5120% because small difficult families receive the same weight as families with many easy generated cases; it does not invalidate the pair-level score.

Failure spread check	Value	Share
Families with at least one Hit@1 failure	1,070	6.35%
Families with at least one Hit@20 failure	0	0.00%
Top ten families’ share of Hit@1 failures	27	2.27%

Table 23: Hit@1 failures occur across 1,070 target families; the ten largest family contributors create 2.27% of all misses. The gap is distributed, not produced by a short list of hard-coded names.

6.1 What The Corrupted Text Can Identify

Algorithm 5 uses unweighted optimal-string-alignment distance, abbreviated OSA, inside its raw edit feature. OSA charges one for insertion, deletion, substitution, or one adjacent swap. For example, **firt** to **FRIT** costs one OSA swap but two Levenshtein substitutions. The catalog scan compares each query with all 17,476 compact family keys.

Catalog geometry under OSA	Cases	Share	A5 H@1	A5 H@20	H@1 misses
Gold is the unique nearest family	61,228	92.41%	99.9771%	100.0000%	14
Gold ties with catalog neighbors	4,210	6.35%	86.4133%	100.0000%	572
Another catalog family is closer	819	1.24%	26.3736%	100.0000%	603

Table 24: A5 reaches 99.9771% Hit@1 on the 61,228 cases where the source target is the only nearest catalog family. Ties and closer real families create 1,175 of the 1,189 strict Hit@1 misses.

Status among A5 Hit@1 misses	Cases	Failure share	Ranking	Retrieval
Gold is the unique nearest family	14	1.18%	14	0
Gold ties with catalog neighbors	572	48.11%	572	0
Another catalog family is closer	603	50.71%	603	0

Table 25: Only 14 failures have a source target that is uniquely nearest under A5’s own raw distance. The other 1,175 are catalog near-collisions or distance ties, although the source family remains the strict benchmark answer.

Status among 1,189 misses	Levenshtein n	Levenshtein share	OSA n	OSA share
Gold is the unique nearest family	15	1.26%	14	1.18%
Gold ties with catalog neighbors	566	47.60%	572	48.11%
Another catalog family is closer	608	51.14%	603	50.71%

Table 26: Levenshtein and OSA disagree on 26 row-level failure labels because OSA counts an adjacent swap as one operation. Both views are reported; OSA is primary for diagnosing A5 because it matches the deployed raw-distance implementation.

System	Unique H@1	Unique H@20	Tie H@1	Tie H@20	Closer H@1	Closer H@20
Exact or prefix match	3.01%	3.01%	3.40%	3.56%	0.00%	0.00%
Exhaustive Levenshtein	99.32%	100.00%	35.56%	97.74%	0.37%	66.79%
Jaro-Winkler	94.42%	99.73%	34.89%	88.08%	8.06%	54.46%
Character 3-gram TF-IDF	80.25%	95.46%	15.89%	49.88%	2.93%	22.34%
RapidFuzz token ratio	68.74%	85.77%	33.11%	87.67%	3.66%	51.53%
Phonetic baseline	73.13%	93.85%	29.43%	73.47%	16.85%	54.46%
Algorithm 1, current app	87.70%	94.79%	41.33%	66.01%	15.14%	36.14%
Algorithm 2, external fast	95.84%	98.69%	56.10%	79.52%	21.98%	50.55%
Algorithm 3, rank fusion	95.39%	98.86%	53.97%	80.83%	20.76%	50.43%
Algorithm 4, family rescue	99.85%	99.93%	75.80%	97.24%	7.69%	80.34%
Algorithm 5, evidence-guided rescue	99.98%	100.00%	86.41%	100.00%	26.37%	100.00%

Table 27: Performance changes sharply with catalog geometry. A5 is strongest on unique and tied targets, but a system cannot infer the generated source family from edit distance when another real family is literally closer.

6.2 The 14 Unique-Nearest Failures

These 14 rows are the clearest algorithm defects because no catalog family has an equal or smaller OSA distance than the verified target.

Observed mechanism	Cases	Share of all A5 misses	Direct meaning
Combined-score inversion	14	1.18%	Expected family is present, uniquely nearest, but has a lower combined score.
Bounded correction override	0	0.00%	A correction moved a lower-score family above the uniquely nearest target.
No candidates returned	0	0.00%	Candidate generation returned an empty list.
Target outside top 20	0	0.00%	Other candidates were returned, but the uniquely nearest target was absent.

Table 28: The 14 unique-nearest failures divide into 14 reranking defects and 0 candidate-generation defects. The groups are exhaustive and mutually exclusive.

Failure level	Cases	4+ bigrams	≤ 4 chars	No result	Pattern
Expected already in top 20	14	8	0	0	Reranking problem: 14 combined-score inversions and 0 bounded-correction overrides.
Expected outside top 20	0	0	0	0	No remaining candidate-generation failures; every verified family is in the top 20.

Table 29: All 14 current unique-nearest failures are ranking misses; no unique-nearest retrieval miss remains. Eight of these rows retain at least four shared bigrams, so candidate order, not missing local evidence, is the residual defect.

Input	Expected	A5 first	Raw d : gold/top	Score: gold/top	Rank	Manual diagnosis
ligncainehydr0chl oride	LIGNOCAINEHYDROCHLORID E	LIDOCAINEHYDROCHLORIDE	2/3	1.8806/2.2115	2	The uniquely nearest target is one edit closer, but the long-name neighbor leads the combined score by 0.3309.
soivinn	SOLVINN	SOLVIN	1/2	0.4585/1.5630	20	Shorter-family evidence overwhelms the full SOLVIN N spelling by 1.1046 score units.
clicynane	DICYNONE	CALCIONATE	3/4	1.0443/1.3082	9	The structural CL-to-D target is one edit closer but remains at rank 9 after bounded corrections.
ivyhuda	IVYBUTA	IVYHONTA	2/3	1.6709/1.7003	2	The uniquely nearest target trails the displayed family by only 0.0294 score units.

Table 30: Fresh A5 reruns expose the internal evidence on retained unique-nearest ranking failures. These examples show why raw-distance advantage can still lose to the combined score.

6.3 Can Another Method Recover The Misses?

Other system	H@1 rescues	Share of A5 misses	H@20 retrievals	Share of A5 misses
Algorithm 2, external fast	203	17.07%	621	52.23%
Algorithm 3, rank fusion	197	16.57%	626	52.65%
Exhaustive Levenshtein	169	14.21%	982	82.59%
Phonetic baseline	149	12.53%	642	53.99%
Jaro-Winkler	148	12.45%	764	64.26%
Algorithm 1, current app	138	11.61%	429	36.08%
RapidFuzz token ratio	135	11.35%	776	65.26%
Character 3-gram TF-IDF	66	5.55%	368	30.95%
Algorithm 4, family rescue	12	1.01%	1,020	85.79%
Exact or prefix match	0	0.00%	5	0.42%

Table 31: Each row asks how the other system handles exactly the cases A5 misses at rank 1. Counts overlap because several systems can recover the same case; exhaustive Levenshtein supplies the largest complementary H@1 set.

Property of A5 retrieval misses that every other system also misses	Cases
Total	0
Mixed-operation rows	0
Effective Levenshtein distance 3–5	0
Another family closer	0
Gold tied nearest	0
Gold unique nearest	0

Table 32: Only 0 current A5 retrieval misses also remain outside every other system’s top 20. A zero count confirms that Algorithm 5 now retrieves every verified family in this clean core.

Nearest-family override rule	Changed	Fixed	Broken	Net	Counterfactual H@1
Any unique nearest family	161	14	125	-111	98.0379%
Only when A5 returns no result	0	0	0	+0	98.2055%
Only queries of at most four characters	13	0	9	-9	98.1919%
At most four characters and at most one bigram	11	0	7	-7	98.1949%

Table 33: A global nearest-family override changes 161 rows, fixes 14 misses, and breaks 125 correct multi-signal decisions, a net change of -111. The no-result-only fallback fixes 0 and breaks 0 because the current algorithm has no empty results.

6.4 Rows Inspected Manually

Table 34: 5 current failures inspected against the full catalog and fresh A5 candidate evidence. Here n is the number of catalog families at the nearest OSA distance; > 20 means absent from the returned top 20, not literal rank 999.

Class	Input	Expected	A5 first	Rank	Gold d	Nearest family, distance, count	What happened
Unique nearest, ranking	soivinn	SOLVINN	SOLVIN	20	1	SOLVIN N ($d=1, n=1$)	SOLVIN N is uniquely nearest at distance 1, but the shorter SOLVIN family receives a much larger family score and pushes it to rank 20.
Nearest-distance tie	conal	COBAL	CONIL	3	1	COBAL; CONAZ; CONIL ($d=1, n=3$)	COBAL and CONIL are both one edit from CONAL, so raw distance cannot decide; A5 ranks COBAL third.

Class	Input	Expected	A5 first	Rank	Gold d	Nearest family, distance, count	What happened
Another family closer	rimunal	REMINYL	NIFUNAL	2	3	NIFUNAL (d=2, n=1)	NIFUNAL is distance 2 while source target REMINYL is distance 3; A5's first result is the literal nearest family.
Another family closer	blaxeedoil	BLACKSEEDOIL	FLAXSEEDOIL	2	3	FLAX SEED OIL (d=2, n=1)	FLAX SEED OIL is distance 2 while BLACK SEED OIL is distance 3, making the generated phonetic corruption a real catalog near-collision.
Another family closer	go5y	GOSAY	GORY	2	2	GORY (d=1, n=1)	GORY is distance 1 while GOSAY is distance 2; the wrong first result follows the stronger visible spelling.

7 Which Errors Cause Failure

System	Overall H@1	1 edit	2 edits	3 edits	4 edits	5 edits
Exact or prefix match	3.00%	5.14%	0.00%	0.00%	0.00%	0.00%
Exhaustive Levenshtein	94.05%	99.04%	91.89%	76.10%	43.47%	64.54%
Jaro-Winkler	89.57%	97.35%	84.24%	64.48%	47.03%	60.70%
Character 3-gram TF-IDF	75.21%	87.96%	62.44%	45.12%	21.14%	34.50%
RapidFuzz token ratio	65.67%	71.22%	62.33%	47.44%	29.22%	30.35%
Phonetic baseline	69.66%	72.60%	68.00%	60.64%	33.97%	51.76%
Algorithm 1, current app	83.86%	92.41%	80.86%	51.41%	13.06%	2.56%
Algorithm 2, external fast	92.41%	96.94%	90.66%	76.95%	37.77%	45.69%
Algorithm 3, rank fusion	91.84%	96.40%	90.36%	75.37%	36.82%	45.05%
Algorithm 4, family rescue	97.19%	99.55%	96.59%	88.29%	59.86%	81.79%
Algorithm 5, evidence-guided rescue	98.21%	99.69%	97.87%	91.73%	80.76%	96.17%

Table 35: Hit@1 by effective Levenshtein count. One edit means one insertion, deletion, or replacement; adjacent swaps still cost two here.

System	Overall H@20	1 edit	2 edits	3 edits	4 edits	5 edits
Exact or prefix match	3.01%	5.16%	0.00%	0.00%	0.00%	0.00%
Exhaustive Levenshtein	99.44%	100.00%	99.98%	96.84%	73.87%	85.30%
Jaro-Winkler	98.43%	99.97%	98.62%	90.23%	83.37%	88.82%
Character 3-gram TF-IDF	91.66%	98.49%	86.82%	71.26%	41.09%	60.06%
RapidFuzz token ratio	85.47%	87.66%	85.00%	76.37%	66.51%	63.58%
Phonetic baseline	92.07%	94.08%	91.60%	84.53%	60.81%	75.40%
Algorithm 1, current app	92.23%	98.05%	94.27%	61.00%	15.68%	4.15%
Algorithm 2, external fast	96.88%	99.87%	96.91%	84.78%	46.79%	48.56%
Algorithm 3, rank fusion	97.12%	99.75%	97.41%	86.34%	46.79%	48.56%
Algorithm 4, family rescue	99.52%	99.90%	99.74%	98.15%	81.95%	89.78%
Algorithm 5, evidence-guided rescue	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%

Table 36: Hit@20 by effective Levenshtein count. The distance-four and distance-five columns expose candidate-generation limits that overall accuracy can hide.

System	Mixed	Visual/OCR	Phonetic	Vowel	Deletion	Swap	Insertion	Keyboard	Other
Exact or prefix match	0.0%	0.0%	0.1%	0.0%	48.9%	0.0%	0.0%	0.0%	0.2%
Exhaustive Levenshtein	84.2%	98.4%	98.9%	95.1%	96.4%	94.0%	99.8%	98.2%	99.0%
Jaro-Winkler	75.7%	95.3%	96.3%	79.2%	98.7%	99.4%	99.5%	92.4%	98.9%
Character 3-gram TF-IDF	51.2%	86.5%	84.4%	57.2%	93.5%	75.2%	97.2%	78.2%	99.4%
RapidFuzz token ratio	56.9%	67.5%	67.2%	64.5%	74.3%	66.0%	80.1%	74.6%	72.2%
Phonetic baseline	66.1%	64.3%	83.1%	78.3%	47.4%	81.2%	74.7%	47.5%	79.0%
Algorithm 1, current app	68.9%	84.9%	95.4%	96.3%	81.3%	94.9%	91.6%	75.7%	93.6%
Algorithm 2, external fast	82.9%	93.1%	97.9%	97.5%	97.8%	98.8%	98.4%	86.8%	99.5%
Algorithm 3, rank fusion	82.2%	93.0%	98.0%	97.4%	92.6%	98.1%	97.8%	87.3%	99.4%
Algorithm 4, family rescue	92.5%	98.5%	99.4%	98.5%	99.5%	99.5%	99.6%	98.1%	99.6%
Algorithm 5, evidence-guided rescue	95.0%	99.2%	99.8%	98.8%	99.6%	100.0%	99.6%	98.5%	100.0%

Table 37: Hit@1 by rule-derived mutation mechanism. Mixed means that category and error labels contain more than one mechanism, such as deletion plus phonetic substitution. Hit@20 is 100% for Algorithm 5 in every mechanism.

What is a bigram? A character bigram is two adjacent compact characters. DAFLON contains DA, AF, FL, LO, and ON. Query 0aflon shares AF, FL, LO, and ON with DAFLON. A zero-bigram row keeps no adjacent two-character anchor.

System	Overall H@1	0 shared	1 shared	2–3 shared	4+ shared
Exact or prefix match	3.00%	0.00%	0.00%	1.28%	3.28%
Exhaustive Levenshtein	94.05%	9.74%	33.22%	71.24%	98.41%
Jaro-Winkler	89.57%	18.05%	29.90%	60.50%	94.55%
Character 3-gram TF-IDF	75.21%	0.00%	0.00%	17.04%	83.90%
RapidFuzz token ratio	65.67%	16.05%	25.35%	59.50%	67.48%
Phonetic baseline	69.66%	21.20%	22.38%	34.92%	74.92%
Algorithm 1, current app	83.86%	26.93%	30.86%	57.77%	88.28%
Algorithm 2, external fast	92.41%	37.82%	39.60%	70.52%	96.32%
Algorithm 3, rank fusion	91.84%	32.38%	36.71%	69.77%	95.84%
Algorithm 4, family rescue	97.19%	40.11%	61.80%	87.61%	99.34%
Algorithm 5, evidence-guided rescue	98.21%	57.02%	74.30%	91.89%	99.66%

Table 38: Hit@1 by shared adjacent-character evidence. The table measures how much local text remains for first-rank ordering; Algorithm 5 still retrieves all groups within the top 20.

Algorithm 5 reaches 57.02% Hit@1 when the query and target share no adjacent pair. This is the strictest first-rank test with almost no local lexical anchor; Hit@20 remains 100%.

For compact query q_c and compact target y_c , normalized distance is

$$r = \frac{L(q_c, y_c)}{\max(|y_c|, 1)}, \quad \text{extreme when } r > 0.60,$$

where L is Levenshtein distance. The threshold means more than 60 edits per 100 target characters, not 60% of the dataset. For example, 2ix→ZEX has two edits over a three-character target, so $r = 2/3 = 0.667$ and the row is extreme.

System	Std. n	Std. H@1	Std. H@20	High n	High H@1	High H@20	Extreme n	Extreme H@1	Extreme H@20
Exact or prefix match	64,664	3.08%	3.09%	1,470	0.00%	0.00%	123	0.00%	0.00%
Exhaustive Levenshtein	64,664	95.83%	99.99%	1,470	23.47%	81.50%	123	0.81%	26.83%
Jaro-Winkler	64,664	91.10%	99.25%	1,470	29.39%	66.94%	123	3.25%	40.65%
Character 3-gram TF-IDF	64,664	76.76%	93.30%	1,470	12.38%	26.60%	123	7.32%	8.13%
RapidFuzz token ratio	64,664	66.72%	86.10%	1,470	24.63%	62.38%	123	6.50%	32.52%
Phonetic baseline	64,664	70.94%	93.02%	1,470	19.12%	56.12%	123	0.81%	20.33%
Algorithm 1, current app	64,664	85.41%	93.66%	1,470	22.65%	36.80%	123	1.63%	5.69%
Algorithm 2, external fast	64,664	93.81%	98.02%	1,470	38.16%	53.95%	123	4.07%	8.94%
Algorithm 3, rank fusion	64,664	93.28%	98.24%	1,470	35.71%	55.03%	123	4.88%	12.20%
Algorithm 4, family rescue	64,664	98.35%	99.87%	1,470	52.99%	89.46%	123	13.82%	31.71%
Algorithm 5, evidence-guided rescue	64,664	98.93%	100.00%	1,470	69.05%	100.00%	123	68.29%	100.00%

Table 39: Recovery by normalized-distance cohort. Standard has $r \leq 0.40$, high has $0.40 < r \leq 0.60$, and extreme has $r > 0.60$.

The extreme cohort contains 123 cases. Algorithm 5 reaches 68.29% Hit@1 and 100.00% Hit@20 on those rows.

Table 40: Distribution and Algorithm 5 recovery inside the extreme cohort. Rows under different dimensions describe the same extreme cases and should not be summed together.

Dimension	Group	Cases	Share	A5 H@1	A5 H@20
Effective edits	2	12	9.76%	100.00%	100.00%
Effective edits	3	9	7.32%	100.00%	100.00%
Effective edits	4	58	47.15%	44.83%	100.00%
Effective edits	5	44	35.77%	84.09%	100.00%
Operation family	Mixed Operations	104	84.55%	62.50%	100.00%
Operation family	Other	9	7.32%	100.00%	100.00%
Operation family	Phonetic	1	0.81%	100.00%	100.00%
Operation family	Transposition	5	4.07%	100.00%	100.00%
Operation family	Visual Or OCR	4	3.25%	100.00%	100.00%
Shared bigrams	0 Shared Bigrams	41	33.33%	51.22%	100.00%
Shared bigrams	1 Shared Bigram	51	41.46%	72.55%	100.00%
Shared bigrams	2 3 Shared Bigrams	31	25.20%	83.87%	100.00%
Difficulty	Easy	2	1.63%	100.00%	100.00%
Difficulty	Extreme	61	49.59%	65.57%	100.00%
Difficulty	Hard	54	43.90%	66.67%	100.00%
Difficulty	Medium	6	4.88%	100.00%	100.00%

Input	Expected	r	Shared bigrams	A5 first	Expected rank	Category
pryozlarn	PRAZOLAM	0.625	2	PRAZOLAM	1	Combined Errors Effective 4 +
lvce	VOLCA	0.800	0	LEV CET	4	Combined Errors Effective 3
milmila	MILA	0.750	3	MILA	1	Speed Typing Errors
rnol	MOL	0.667	1	MOL	1	Ligature Confusion
ivee	IVY	0.667	1	IVY	1	Multi Char Phonetic Confusion
pb5	BBS	0.667	0	BBS	1	OCR + Error Effective 2
2ix	ZEX	0.667	0	ZEX	1	OCR + Error Effective 2
nta	NAT	0.667	0	NAT	1	Transposition Position Weighted
alx	LAX	0.667	0	LAX	1	Transposition Position Weighted
ful	FLU	0.667	0	FLU	1	Transposition Position Weighted

Table 41: Concrete extreme rows show what information survives, what A5 returns, and whether the verified family remains inside the top 20.

Table 42: Hit@20 by clean category for all six classical baselines.
This exposes which mutation mechanisms suit each single-signal method.

Clean category	n	Exact/prefix	Leven.	Jaro-W.	3-gram	Token	Phonetic
Single Letter Visual Confusion	7,841	0.0%	100.0%	100.0%	98.7%	86.4%	91.7%
Combined Errors Effective 2	7,463	0.0%	99.5%	98.1%	75.3%	85.4%	93.6%
Single Letter Phonetic Confusion	5,983	0.0%	100.0%	99.9%	98.1%	85.5%	100.0%
Multi Char Phonetic Confusion	4,991	0.3%	100.0%	99.9%	98.6%	86.8%	95.4%
Single Char Deletion Position Weighted	4,966	39.8%	100.0%	100.0%	99.2%	87.9%	82.6%
Wrong Vowels Relabelled By Actual Changes	4,870	0.0%	99.9%	97.0%	84.2%	85.2%	100.0%
Combined Errors Effective 3	4,346	0.0%	94.3%	86.1%	65.3%	73.9%	77.8%
OCR Letter Digit Confusion	3,998	0.0%	100.0%	100.0%	98.9%	87.3%	93.1%
Ligature Confusion	3,992	0.0%	100.0%	99.4%	97.4%	84.1%	78.7%
Transposition Position Weighted	3,000	0.0%	100.0%	99.8%	94.9%	85.1%	95.7%
OCR + Error Effective 2	2,997	0.0%	100.0%	99.5%	88.0%	82.3%	97.9%
Single Char Insertion	2,988	0.0%	100.0%	100.0%	99.7%	90.7%	98.8%
Double Letter Reduction Expansion	2,987	0.3%	100.0%	100.0%	99.9%	88.5%	100.0%
Speed Typing Errors	2,981	0.0%	100.0%	100.0%	99.8%	90.7%	94.6%
Keyboard Adjacent Sampled	1,924	0.0%	100.0%	99.9%	94.4%	88.2%	80.9%
Combined Errors Effective 4 +	563	0.0%	86.0%	83.7%	54.9%	63.4%	68.7%
Combined Errors Effective 1	496	0.4%	100.0%	100.0%	94.0%	91.3%	95.4%
Dangerous Name Pair Gold Closer	158	0.0%	100.0%	100.0%	80.4%	92.4%	97.5%
Consonant Frame Wrong Vowels Heavy	68	0.0%	100.0%	91.2%	66.2%	80.9%	100.0%
Extreme Combined Effective 2	4	0.0%	100.0%	100.0%	50.0%	100.0%	100.0%
Extreme Combined Effective 3	3	0.0%	100.0%	66.7%	66.7%	33.3%	100.0%
OCR + Error Effective 1	2	0.0%	100.0%	100.0%	50.0%	100.0%	100.0%

Table 43: Hit@1 and Hit@20 by clean category for Algorithms 1–5.
Each row reports the same category memberships and cases for all systems.

Clean category	n	A1 H@1	A2 H@1	A3 H@1	A4 H@1	A5 H@1	A1 H@20	A2 H@20	A3 H@20	A4 H@20	A5 H@20
Single Letter Visual Confusion	7,841	88.2%	94.5%	94.4%	99.6%	99.7%	97.3%	99.9%	99.8%	100.0%	100.0%
Combined Errors Effective 2	7,463	77.8%	87.1%	86.1%	95.3%	96.1%	92.0%	95.7%	96.0%	99.7%	100.0%
Single Letter Phonetic Confusion	5,983	98.8%	97.4%	98.2%	99.6%	99.8%	99.6%	99.9%	99.9%	99.9%	100.0%
Multi Char Phonetic Confusion	4,991	91.3%	98.5%	97.7%	99.2%	99.8%	97.7%	99.8%	99.7%	99.9%	100.0%
Single Char Deletion Position Weighted	4,966	86.7%	98.4%	94.8%	99.6%	99.7%	95.7%	99.9%	99.8%	100.0%	100.0%
Wrong Vowels Relabelled By Actual Changes	4,870	96.3%	97.5%	97.5%	98.5%	98.8%	98.7%	99.0%	99.1%	99.5%	100.0%

Clean category	<i>n</i>	A1 H@1	A2 H@1	A3 H@1	A4 H@1	A5 H@1	A1 H@20	A2 H@20	A3 H@20	A4 H@20	A5 H@20
Combined Errors Effective 3	4,346	34.1%	64.4%	62.6%	81.5%	86.7%	45.5%	75.6%	77.9%	96.4%	100.0%
OCR Letter Digit Confusion	3,998	94.2%	96.9%	96.7%	99.5%	99.8%	98.9%	99.8%	99.9%	99.8%	100.0%
Ligature Confusion	3,992	69.1%	86.5%	86.5%	95.2%	97.7%	94.5%	97.4%	98.1%	99.9%	100.0%
Transposition Position Weighted	3,000	94.9%	98.8%	98.1%	99.5%	100.0%	98.4%	99.8%	99.8%	99.8%	100.0%
OCR + Error Effective 2	2,997	81.9%	91.4%	92.5%	97.2%	99.2%	92.1%	94.3%	96.0%	99.5%	100.0%
Single Char Insertion	2,988	91.8%	98.7%	98.0%	99.7%	99.7%	99.0%	100.0%	99.8%	100.0%	100.0%
Double Letter Reduction Expansion	2,987	98.8%	99.9%	99.8%	99.9%	100.0%	99.9%	99.9%	99.9%	99.9%	100.0%
Speed Typing Errors	2,981	86.6%	98.6%	97.4%	99.3%	99.9%	90.2%	99.8%	99.8%	100.0%	100.0%
Keyboard Adjacent Sampled	1,924	75.7%	86.8%	87.2%	98.1%	98.5%	92.7%	99.6%	98.7%	99.6%	100.0%
Combined Errors Effective 4 +	563	2.0%	39.3%	38.5%	73.5%	94.1%	3.0%	45.8%	45.6%	88.1%	100.0%
Combined Errors Effective 1	496	91.5%	96.4%	94.8%	99.8%	99.8%	98.6%	100.0%	100.0%	100.0%	100.0%
Dangerous Name Pair Gold Closer	158	82.9%	87.3%	85.4%	97.5%	98.1%	91.8%	98.7%	98.1%	100.0%	100.0%
Consonant Frame Wrong Vowels Heavy	68	91.2%	97.1%	95.6%	97.1%	97.1%	97.1%	100.0%	100.0%	100.0%	100.0%
Extreme Combined Effective 2	4	50.0%	75.0%	75.0%	75.0%	75.0%	75.0%	100.0%	75.0%	100.0%	100.0%
Extreme Combined Effective 3	3	66.7%	66.7%	66.7%	66.7%	66.7%	66.7%	66.7%	66.7%	100.0%	100.0%
OCR + Error Effective 1	2	50.0%	100.0%	50.0%	100.0%	100.0%	50.0%	100.0%	100.0%	100.0%	100.0%

Table 44: Algorithm 5 Hit@1 failure impact by clean category. Membership categories can overlap on rare rows, so shares are not summed. Combined Errors Effective 3 creates the largest absolute first-rank miss count, 576 cases; this need not be the highest within-category failure rate.

Clean category	Cases	H@1	H@20	Misses	Failure rate	Failure share
Combined Errors Effective 3	4,346	86.75%	100.00%	576	13.25%	48.44%
Combined Errors Effective 2	7,463	96.07%	100.00%	293	3.93%	24.64%
Ligature Confusion	3,992	97.72%	100.00%	91	2.28%	7.65%
Wrong Vowels Relabelled By Actual Changes	4,870	98.79%	100.00%	59	1.21%	4.96%
Combined Errors Effective 4 +	563	94.14%	100.00%	33	5.86%	2.78%
Keyboard Adjacent Sampled	1,924	98.54%	100.00%	28	1.46%	2.35%
Single Letter Visual Confusion	7,841	99.69%	100.00%	24	0.31%	2.02%
OCR + Error Effective 2	2,997	99.23%	100.00%	23	0.77%	1.93%
Single Char Deletion Position Weighted	4,966	99.66%	100.00%	17	0.34%	1.43%
Single Letter Phonetic Confusion	5,983	99.80%	100.00%	12	0.20%	1.01%
Multi Char Phonetic Confusion	4,991	99.80%	100.00%	10	0.20%	0.84%
OCR Letter Digit Confusion	3,998	99.77%	100.00%	9	0.23%	0.76%
Single Char Insertion	2,988	99.70%	100.00%	9	0.30%	0.76%
Speed Typing Errors	2,981	99.87%	100.00%	4	0.13%	0.34%
Dangerous Name Pair Gold Closer	158	98.10%	100.00%	3	1.90%	0.25%

Clean category	Cases	H@1	H@20	Misses	Failure rate	Failure share
Consonant Frame Wrong Vowels Heavy	68	97.06%	100.00%	2	2.94%	0.17%
Combined Errors Effective 1	496	99.80%	100.00%	1	0.20%	0.08%
Extreme Combined Effective 2	4	75.00%	100.00%	1	25.00%	0.08%
Extreme Combined Effective 3	3	66.67%	100.00%	1	33.33%	0.08%
Transposition Position Weighted	3,000	100.00%	100.00%	0	0.00%	0.00%
Double Letter Reduction Expansion	2,987	100.00%	100.00%	0	0.00%	0.00%
OCR + Error Effective 1	2	100.00%	100.00%	0	0.00%	0.00%

8 Danger And Equal-Distance Cases

Table 45: Performance on 158 gold-closer dangerous pairs. The query is closer to the expected family than the named competitor, so these remain scoreable but need cautious presentation.

System	Danger cases	H@1	H@20	Unsafe confident top-1
Exact or prefix match	158	0.00%	0.00%	0.00%
Exhaustive Levenshtein	158	86.71%	100.00%	0.00%
Jaro-Winkler	158	81.01%	100.00%	0.00%
Character 3-gram TF-IDF	158	35.44%	80.38%	0.00%
RapidFuzz token ratio	158	75.95%	92.41%	0.00%
Phonetic baseline	158	43.04%	97.47%	0.00%
Algorithm 1, current app	158	82.91%	91.77%	0.00%
Algorithm 2, external fast	158	87.34%	98.73%	0.63%
Algorithm 3, rank fusion	158	85.44%	98.10%	0.00%
Algorithm 4, family rescue	158	97.47%	100.00%	0.00%
Algorithm 5, evidence-guided rescue	158	98.10%	100.00%	0.00%

Input	Expected	Recorded competitor	Gold edits	Competitor edits	Why it remains
casty	CASTO	CASHO	1	2	Gold family is strictly closer, so one expected answer remains defensible.
solvyn	SOLVIN	SOLVIA	1	2	Gold family is strictly closer, so one expected answer remains defensible.
levy	LEVA	LECA	1	2	Gold family is strictly closer, so one expected answer remains defensible.
vtop	VITOP	VITON	1	2	Gold family is strictly closer, so one expected answer remains defensible.
imix	EMIX	EMUX	1	2	Gold family is strictly closer, so one expected answer remains defensible.
mopram	MEPRAM	DEPRAM	1	2	Gold family is strictly closer, so one expected answer remains defensible.
lmex	LOMEX	TOMEX	1	2	Gold family is strictly closer, so one expected answer remains defensible.
rlly	ROLLY	JOLLY	1	2	Gold family is strictly closer, so one expected answer remains defensible.

Table 46: Concrete dangerous rows. Unlike an exact real-drug collision, the expected family has strictly smaller edit distance than the alternative.

System	Wrong top-1 with result	Equal-distance wrong top-1	Share	Rank-only ties	Retrieval ties
Exact or prefix match	146	25	17.12%	7	18
Exhaustive Levenshtein	3,943	2,957	74.99%	2,933	24
Jaro-Winkler	6,911	1,754	25.38%	1,409	345
Character 3-gram TF-IDF	16,238	1,317	8.11%	668	649
RapidFuzz token ratio	22,745	2,131	9.37%	1,716	415
Phonetic baseline	20,104	324	1.61%	201	123
Algorithm 1, current app	9,412	566	6.01%	304	262
Algorithm 2, external fast	4,931	867	17.58%	490	377
Algorithm 3, rank fusion	5,383	841	15.62%	514	327
Algorithm 4, family rescue	1,834	934	50.93%	855	79
Algorithm 5, evidence-guided rescue	1,189	563	47.35%	563	0

Table 47: A tie means the wrong first family and verified family have the same raw Levenshtein distance from the query. Equal distance identifies ambiguity, not the correct winner.

Input	Expected	A5 first	Tie distance	Expected rank	Category	Failure level
coex	COPEX	COGEX	1	2	Combined Errors Effective 1	ranking
mepo	MEBO	MEPRO	1	2	Dangerous Name Pair Gold Closer	ranking
roaix	ROSIX	ROMIX	1	2	Keyboard Adjacent Sampled	ranking
ultramdx	ULTRAMOX	ULTRAMIX	1	2	OCR Letter Digit Confusion	ranking
oriana	SORIANA	ORIANNA	1	2	Single Char Deletion Position Weighted	ranking
bnohair	NOHAIR	BIOHAIR	1	3	Single Char Insertion	ranking
opental	PENTAL	ORENTAL	1	2	Single Char Insertion	ranking
iotal	OTAL	TOTAL	1	3	Single Char Insertion	ranking
rollyt	ROLYT	ROLLY	1	2	Single Char Insertion	ranking
varioc	VARIC	VARIO	1	2	Single Char Insertion	ranking

Table 48: Actual Algorithm 5 equal-distance failures from the clean core. Ranking failures already contain the target in the top 20; retrieval failures do not.

Order	Equal-distance handling	Operational rule
1	Keep the complete A5 score order	The score combines raw and weighted edits, position, edge, n-gram, skeleton, phonetic, length, family, and retriever agreement.
2	Permit only bounded corrections	A candidate moves first only when a documented correction has enough independent evidence and a limited score gap.
3	Mark the response ambiguous	Equal raw distance does not justify hidden certainty; show alternatives.
4	Ask for visible evidence	Use strength, form, route, or unreadable-position information to select the family or variant.

Table 49: The ranking policy is evidence aggregation plus clarification, not first-letter or alphabetic priority.

9 Concrete Inputs And Outputs

Comparison	Input	Expected	Other top-1	Other rank	Why A5 adds value
Exact or prefix match	clarenumhairconditioner	DERMENUMHAIRCONDITIONER		> 20	A5 ranks the verified family first at 5 edits; Combined Errors Effective 4 +.
Exhaustive Levenshtein	rivisrpire	RIVAROSPIRE	IVYSPIR	3	A5 ranks the verified family first at 4 edits; Combined Errors Effective 3.
Jaro-Winkler	erxa	EURAX	SEROXAT	12	A5 ranks the verified family first at 3 edits; Combined Errors Effective 2.
Character 3-gram TF-IDF	delusav	DOLOSEV	CALUSA	> 20	A5 ranks the verified family first at 3 edits; Consonant Frame Wrong Vowels Heavy.
RapidFuzz token ratio	forsurehairconditioning	FORSUREHAIRCONDITIONING	MOSHINGHAIR	> 20	A5 ranks the verified family first at 3 edits; Multi Char Phonetic Confusion.
Phonetic baseline	vitvitamine	VITAMINE	BETAVIDONE	> 20	A5 ranks the verified family first at 3 edits; Speed Typing Errors.
Algorithm 1, current app	haurvevu	HAIRVIVE	HAIRBACK	4	A5 ranks the verified family first at 3 edits; Wrong Vowels Relabelled By Actual Changes.
Algorithm 2, external fast	vitrolieal	VITROHEAL	OVITRELLE	2	A5 ranks the verified family first at 2 edits; Ligature Confusion.
Algorithm 3, rank fusion	xoal	XOLA	COOL	5	A5 ranks the verified family first at 2 edits; Dangerous Name Pair Gold Closer.
Algorithm 4, family rescue	Opinov	OPONOV	EPINOR	2	A5 ranks the verified family first at 2 edits; OCR + Error Effective 2.

Table 50: One real A5-only Hit@1 case against each comparison system when such a case exists. These rows expose complementary retrieval rather than average accuracy alone.

Table 51: Up to one ranking failure and one retrieval failure per system, selected deterministically when that failure type exists in the canonical row-level output.

System	Failure	Input	Expected	Top-1	Rank	Edits	Category
Exact or prefix match	ranking	cefzi	CEFZIM	CEFZIL	2	1	Single Char Deletion Position Weighted

System	Failure	Input	Expected	Top-1	Rank	Edits	Category
Exact or prefix match	retrieval	zrttil	ROZTEL		> 20	4	Combined Errors Effective 3
Exhaustive Levenshtein	ranking	tarivn	TARIVAN	TARION	2	1	Combined Errors Effective 1
Exhaustive Levenshtein	retrieval	zrttil	ROZTEL	ACNIL	> 20	4	Combined Errors Effective 3
Jaro-Winkler	ranking	easyrezt	EASYREST	EASYRECT	2	1	Combined Errors Effective 1
Jaro-Winkler	retrieval	liasat	HISTA	LIPOVAST	> 20	4	Combined Errors Effective 3
Character 3-gram TF-IDF	ranking	pschodal	PSYCHODAL	ARCHODAL	2	1	Combined Errors Effective 1
Character 3-gram TF-IDF	retrieval	zrttil	ROZTEL	CORTILON	> 20	4	Combined Errors Effective 3
RapidFuzz token ratio	ranking	easyrezt	EASYREST	EASYRECT	2	1	Combined Errors Effective 1
RapidFuzz token ratio	retrieval	zrttil	ROZTEL	EZAPRIL	> 20	4	Combined Errors Effective 3
Phonetic baseline	ranking	hisvonic	HISTONIC	HEPAFENCE	2	1	Combined Errors Effective 1
Phonetic baseline	retrieval	zrttil	ROZTEL	APIFORTYL	> 20	4	Combined Errors Effective 3
Algorithm 1, current app	ranking	pflox	PEFLOX	PELOX	2	1	Combined Errors Effective 1
Algorithm 1, current app	retrieval	zrttil	ROZTEL	ZEROL	> 20	4	Combined Errors Effective 3
Algorithm 2, external fast	ranking	exludon	EXLUTON	OXELADINE	2	1	Combined Errors Effective 1
Algorithm 2, external fast	retrieval	zrttil	ROZTEL	SUREPTIL	> 20	4	Combined Errors Effective 3
Algorithm 3, rank fusion	ranking	exludon	EXLUTON	OXELADINE	2	1	Combined Errors Effective 1
Algorithm 3, rank fusion	retrieval	zrttil	ROZTEL	SARTALESS	> 20	4	Combined Errors Effective 3
Algorithm 4, family rescue	ranking	vidop	VITOP	VIDROP	2	1	Dangerous Name Pair Gold Closer
Algorithm 4, family rescue	retrieval	zrttil	ROZTEL	SORTI	> 20	4	Combined Errors Effective 3
Algorithm 5, evidence-guided rescue	ranking	rocare	RECARÉ	PROCARE	3	1	Dangerous Name Pair Gold Closer

10 Engineering Decision

Order	Action	Evidence in this report	Acceptance condition
P0	Adopt the clean core as the primary synthetic retrieval file	Every one of 66,257 rows is unique, catalog-resolvable, and scoreable.	Future systems run all case IDs with no silent skips.
P1	Improve the highest-impact A5 failure groups	Table 44 separates rate from total miss contribution.	Raise holdout H@20 without increasing unsafe confidence.
P2	Preserve ambiguity for equal raw distance	Table 47 shows ties remain common among wrong first results.	Alternatives remain visible; no first-character hard code.
P3	Validate product scope and dangerous alternatives	Every row is currently marked product-scope unverified.	A domain reviewer confirms target and alternative identities against the deployment catalog.
P4	Keep OCR and human-study claims separate	This file contains generated text queries, not prescription images or pharmacist decisions.	OCR and user-study results retain their own denominators.

Table 52: Ordered engineering work derived from the measured failure distribution.